

1 Transforming Agriculture With AI-Powered Smart Camera Pesticide Sprayers

2 This Startup's 2kW Batteries Can Be Charged 20% To 80% In 20 Minutes

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1 TRANSFORMING AGRICULTURE WITH AI-POWERED SMART CAMERA PESTICIDE SPRAYERS

To counter uneven distribution, soil contamination, and health risks linked to traditional pesticide spraying methods, Niqo Robotics has developed a precise pesticide application method that uses smart cameras and AI



Niqo RoboSpray

While it is impossible to deny the importance of pesticides in cultivating healthy produce, the environmental impact of uncontrolled and indiscriminate blanket spraying is deeply concerning, leading to uneven distribution, soil contamination, and health risks. Bengaluru-based startup Niqo Robotics has developed an innovative solution to replace traditional spraying methods that involve human-operated knapsack sprayers. Instead, they harness the power of smart cameras and AI for precise

pesticide application in agriculture.

Founder Jaisimha Rao was bitten by the entrepreneurial bug while in New York. He left behind a comfortable life to return to Bengaluru and immerse himself in his family's coffee plantations. It was during this time that he recognised a substantial technological gap within the Indian agricultural sector, which encompasses farmers with varying economic statuses. This observation ignited the idea of catering to a customer segment that values technology and is willing to pay for

solutions to existing problems.

The startup has developed a Niqo RoboSpray, transitioning from a drone-based approach to focusing on ground-based sprayer robots retrofitted with a smart camera system for precision spraying.

The company's design comprises a modular camera unit combined with AI processing and output. "The Niqo RoboSpray is equipped with a smart boom, a special nozzle control system, an agricultural camera and a dashboard to allow the operator to select the spraying mechanism. The sprayer boom platform is the foundation for the nozzle setup and camera installation. We leverage this sprayer boom and integrate it into existing tractors or sprayers through retrofitting," explains Rao.

The camera system integrates several critical components, including an image sensor, FPGA, and the Nvidia AI chip. "The system comprises approximately five to six PCB design boards, which aid in executing intricate timing-based functions, such as image acquisition, comprehensive image analysis, AI-driven computations, and output generation. The camera system must operate synchronously with a player moving through a field at